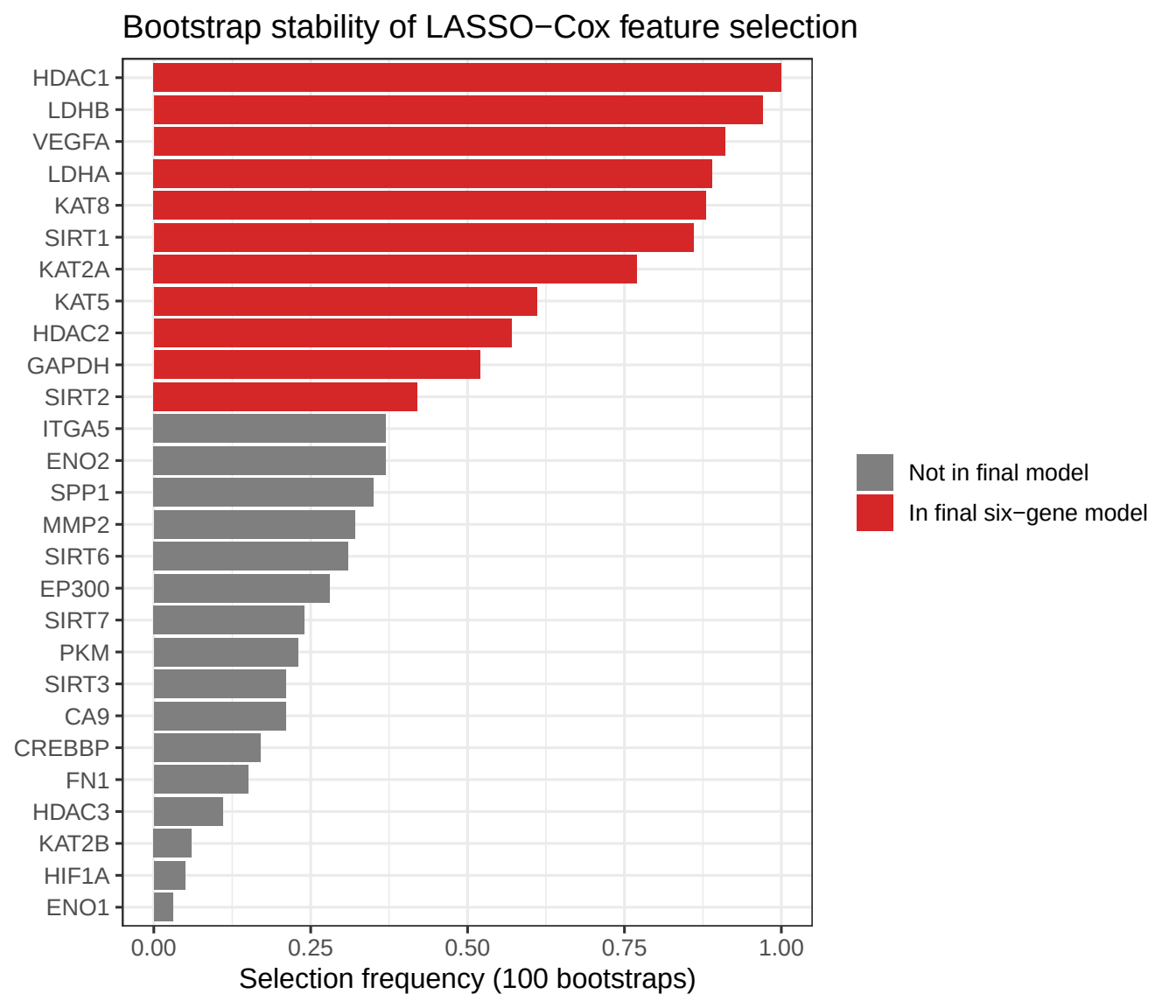
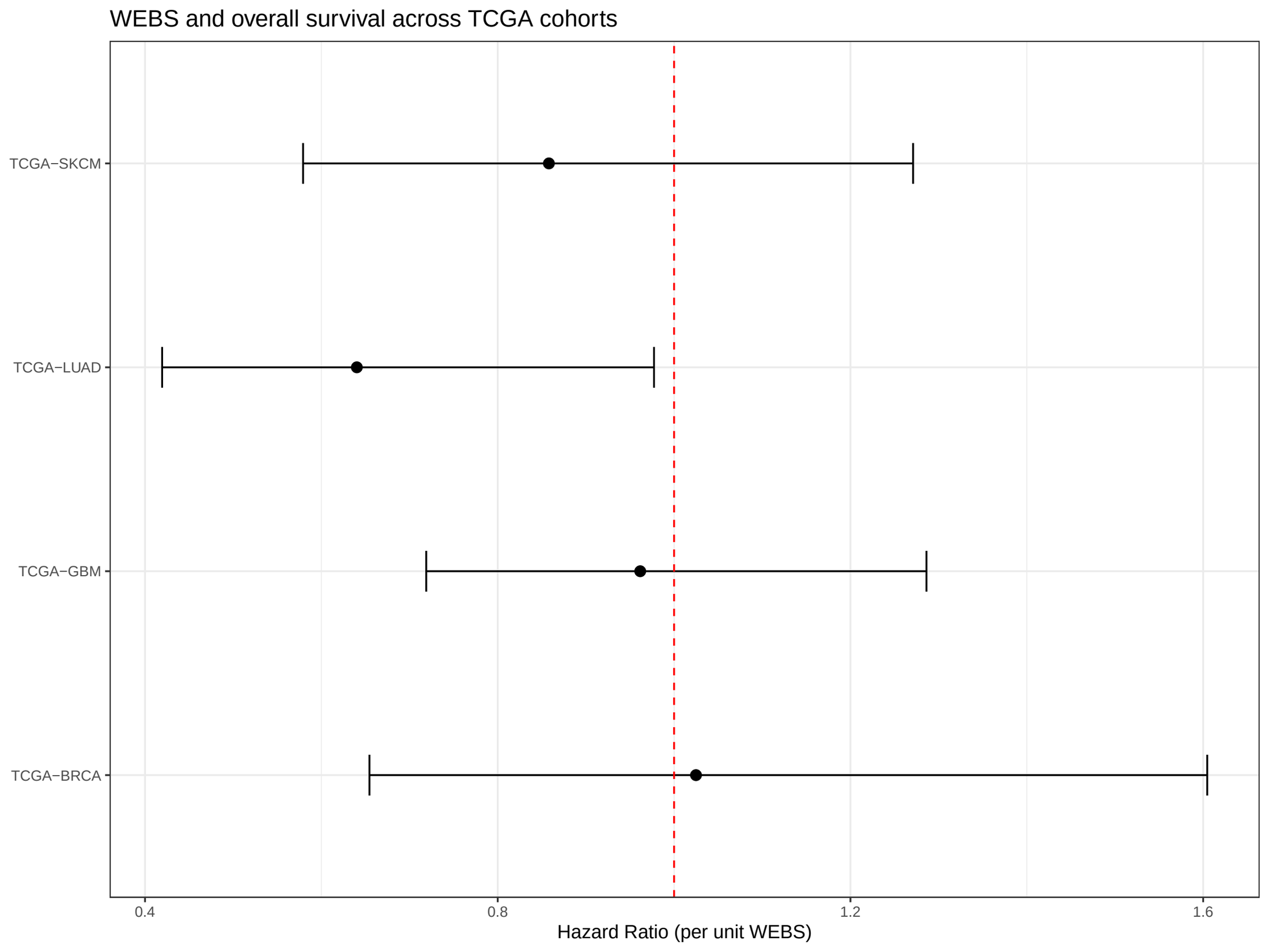
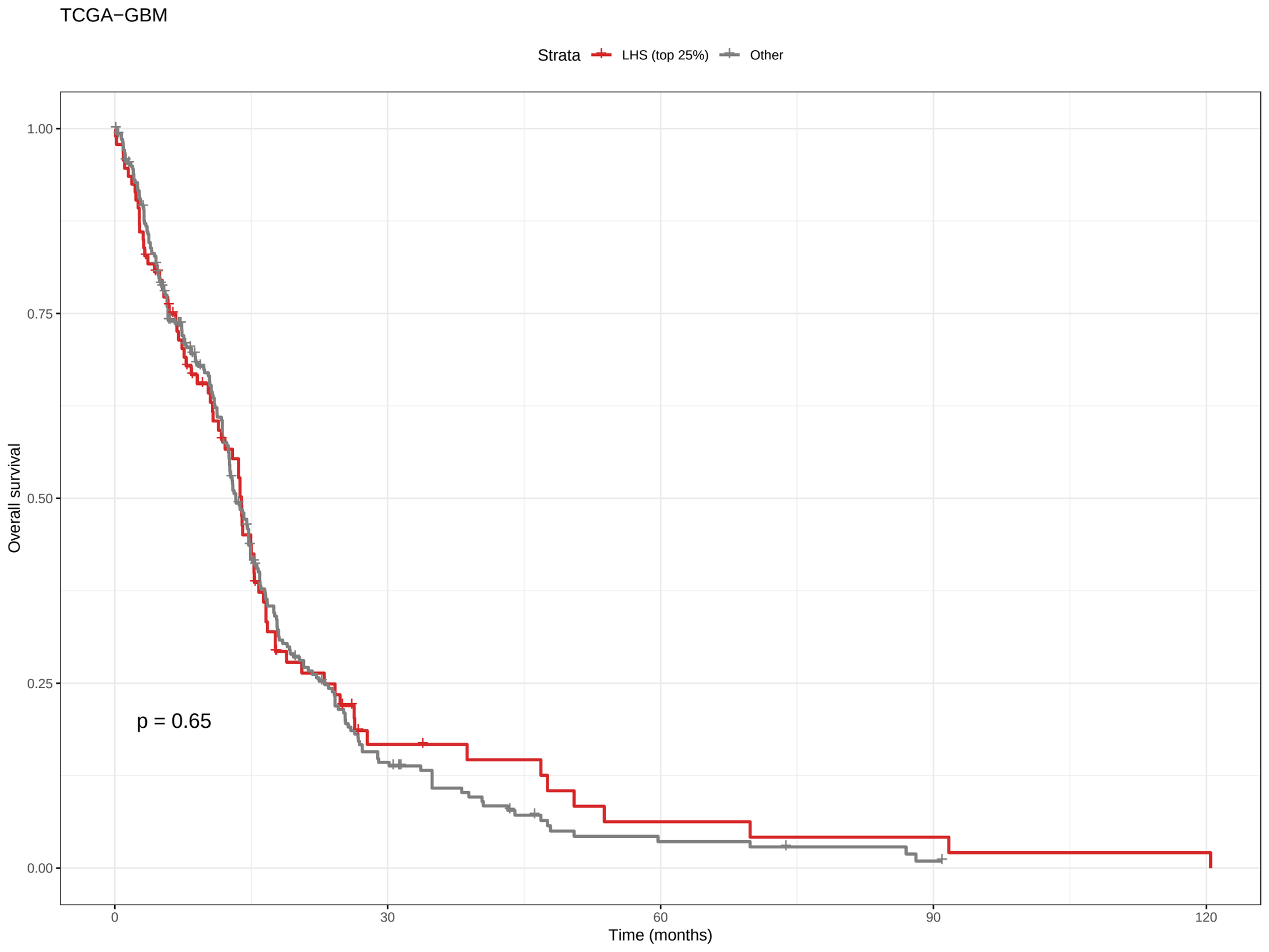
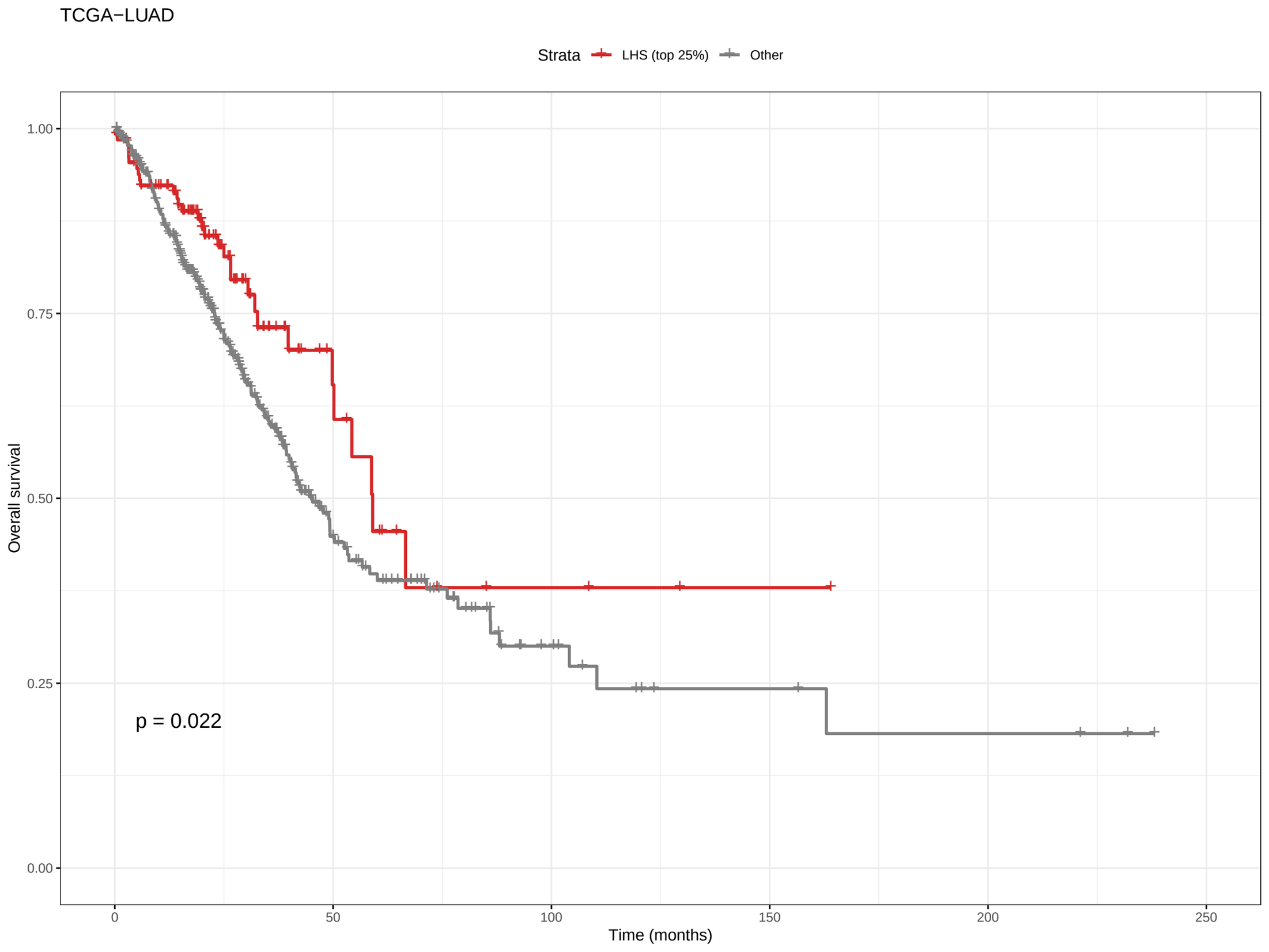
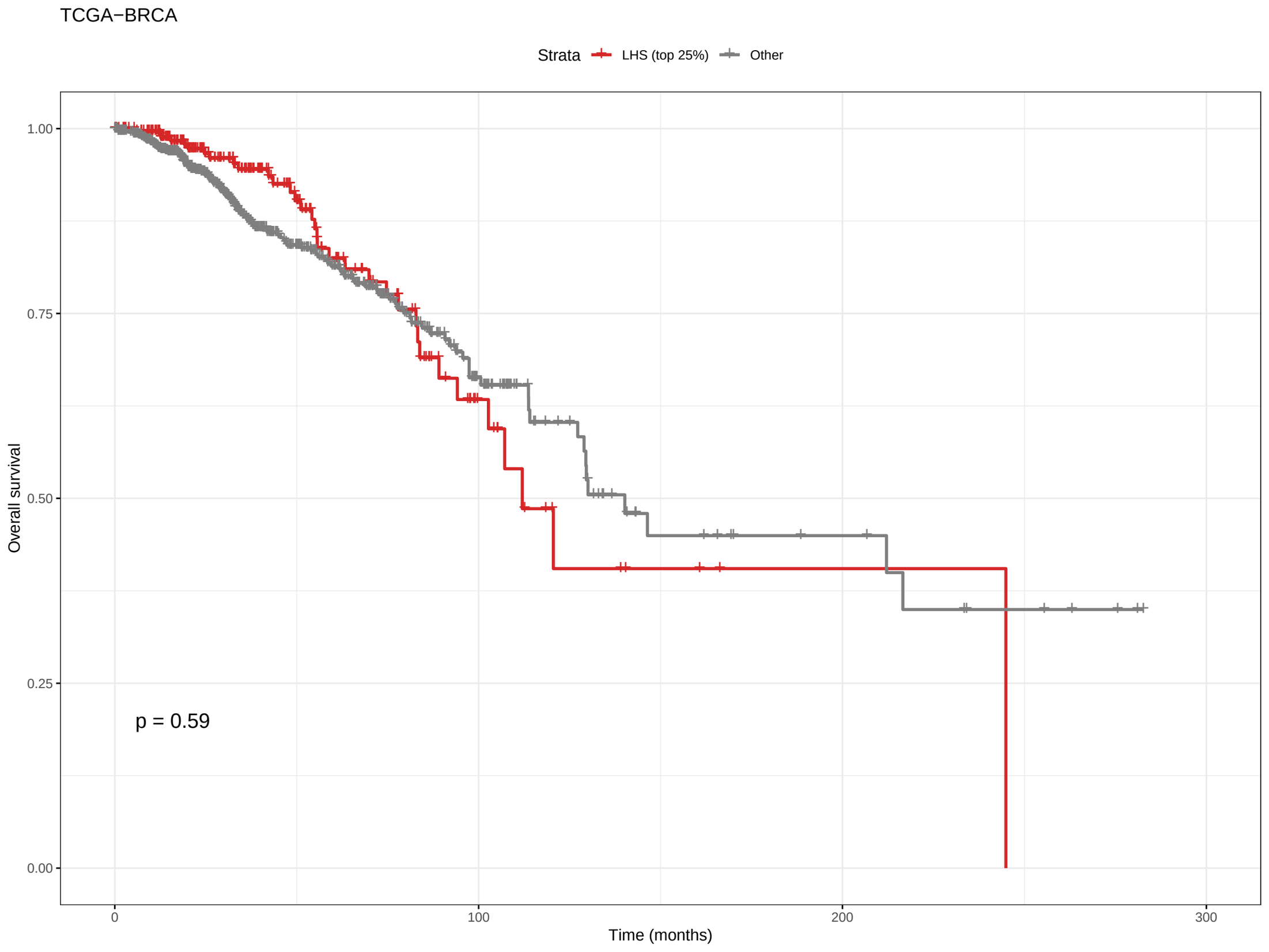
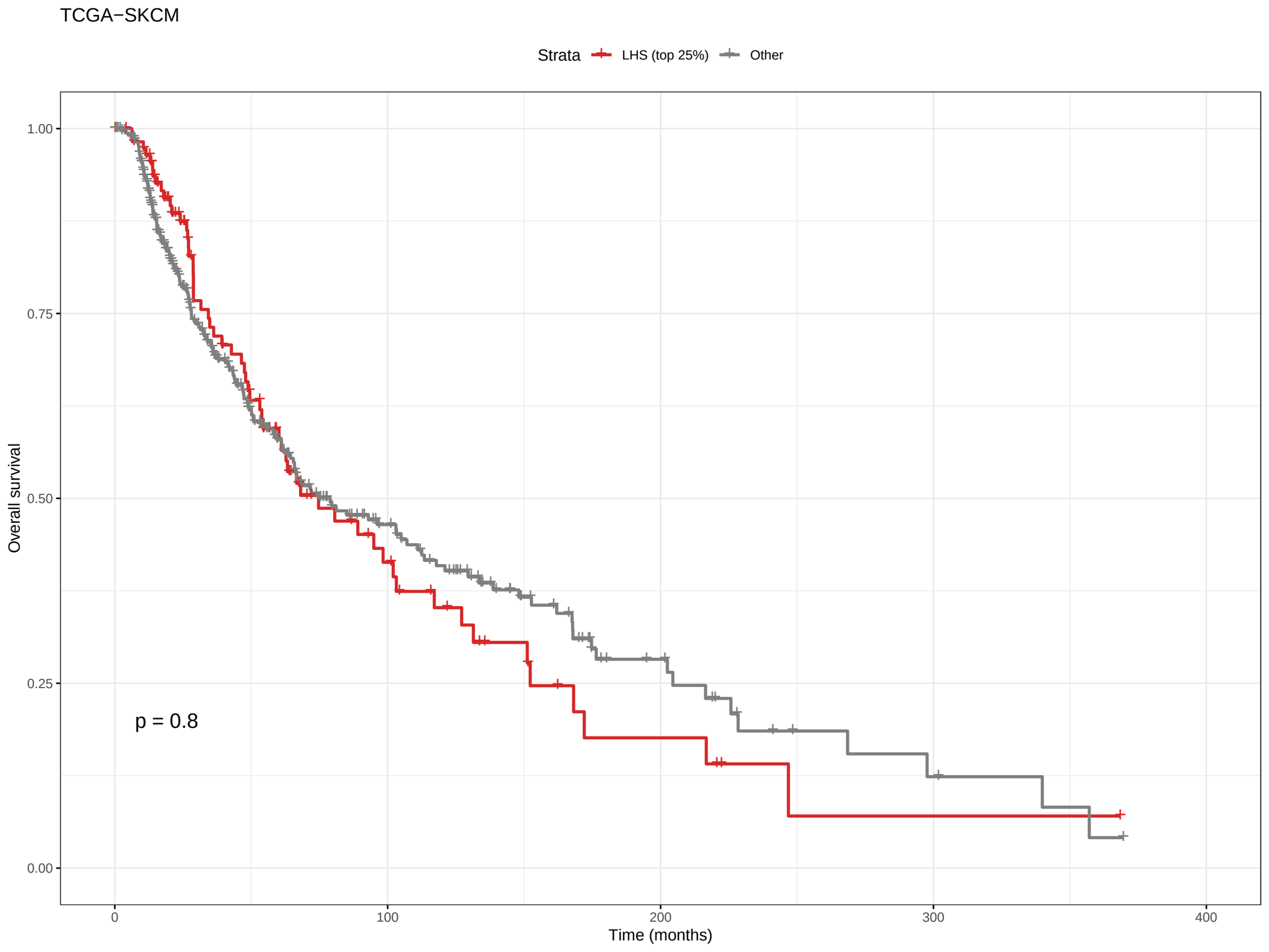

Bootstrap stability of LASSO−Cox feature selection
HDAC1
LDHB
VEGFA
LDHA
KAT8
SIRT1
KAT2A
KAT5
HDAC2
GAPDH
SIRT2
ITGA5
ENO2
SPP1
MMP2
SIRT6
EP300
SIRT7
PKM
SIRT3
CA9
CREBBP
FN1
HDAC3
KAT2B
HIF1A
ENO1
0.00
0.25
0.50
0.75
1.00
Selection frequency (100 bootstraps)
Not in final model
In final six−gene model

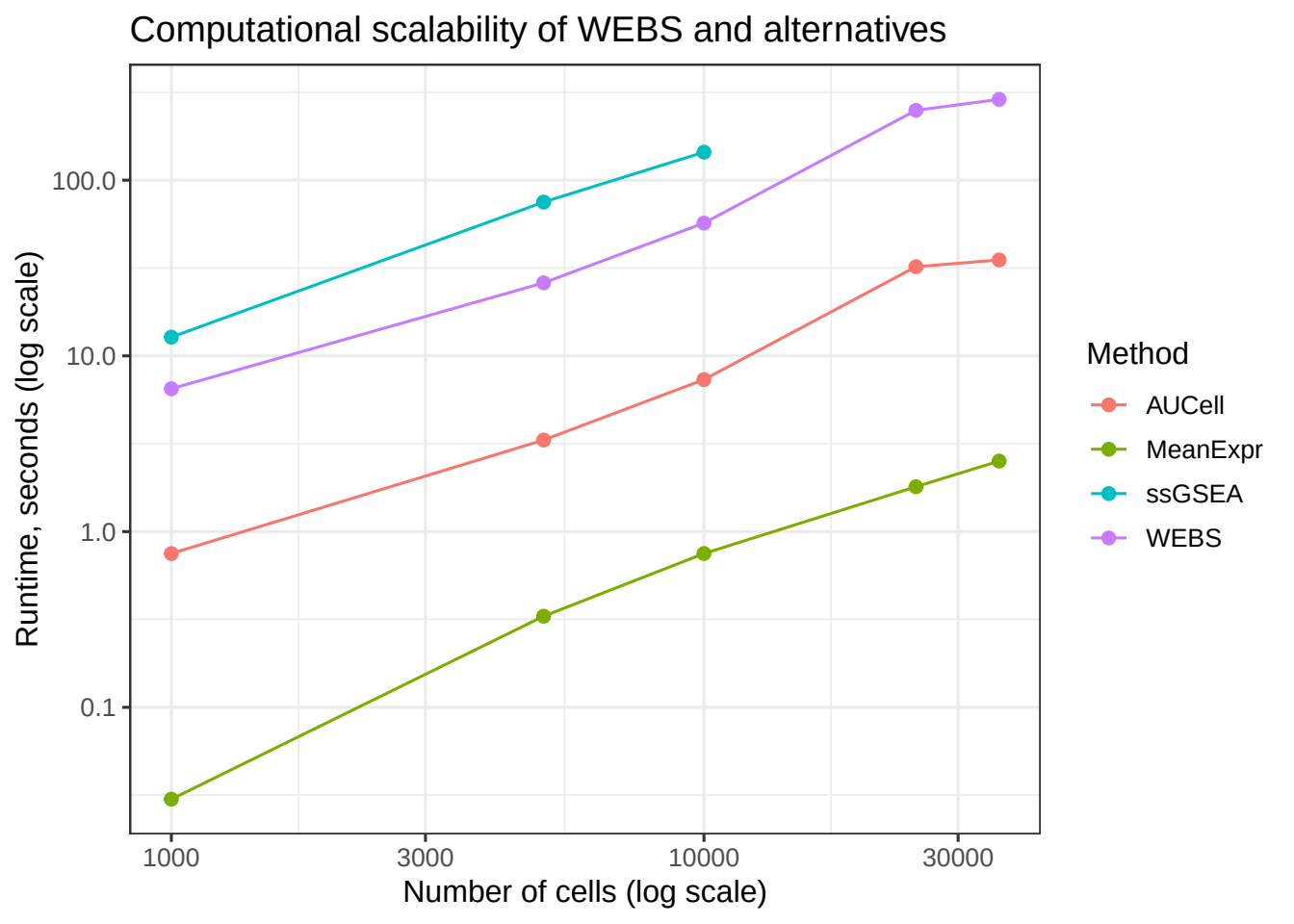

WEBS and overall survival across TCGA cohorts
TCGA−SKCM
TCGA−LUAD
TCGA−GBM
TCGA−BRCA
0.4
0.8
1.2
1.6
Hazard Ratio (per unit WEBS)

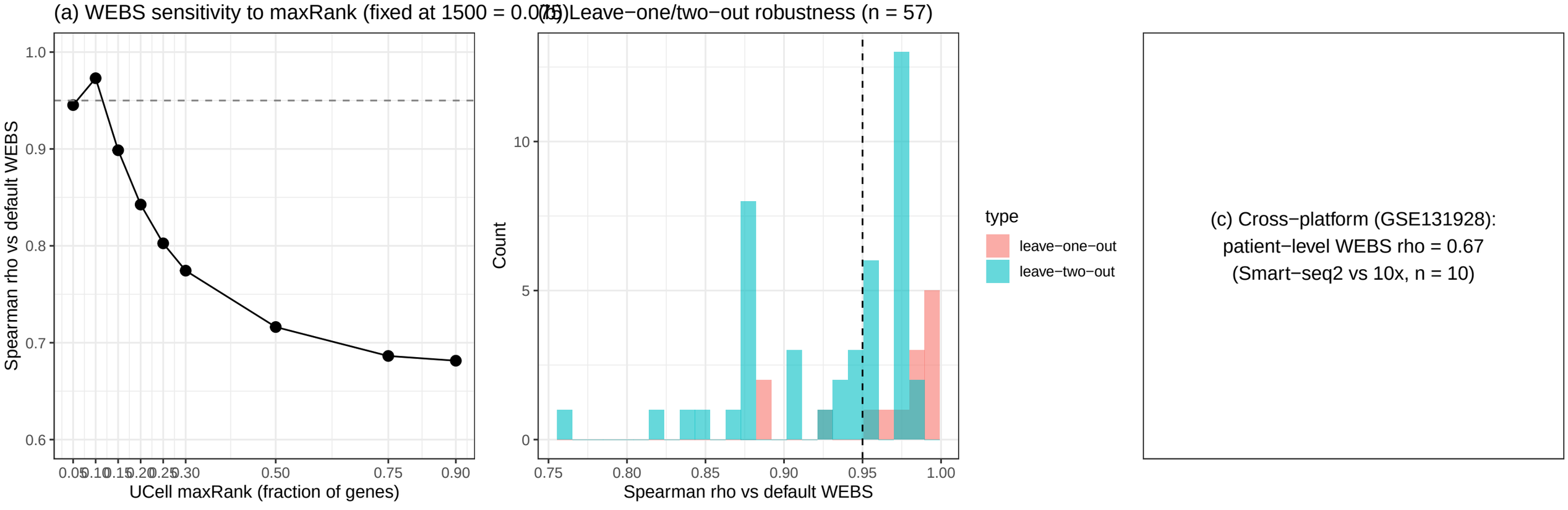

TCGA−GBM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.65
0
30
60
90
120
Time (months)

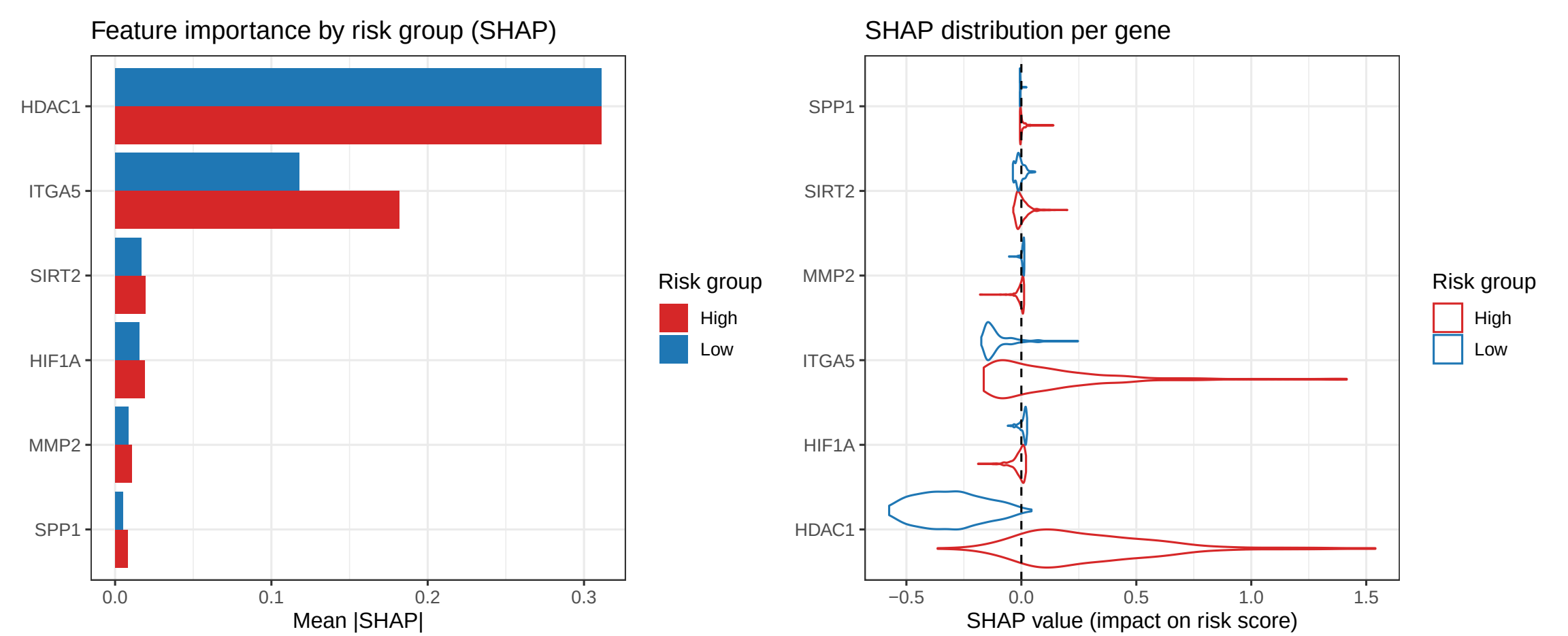

TCGA−LUAD
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.022
0
50
100
150
200
250
Time (months)

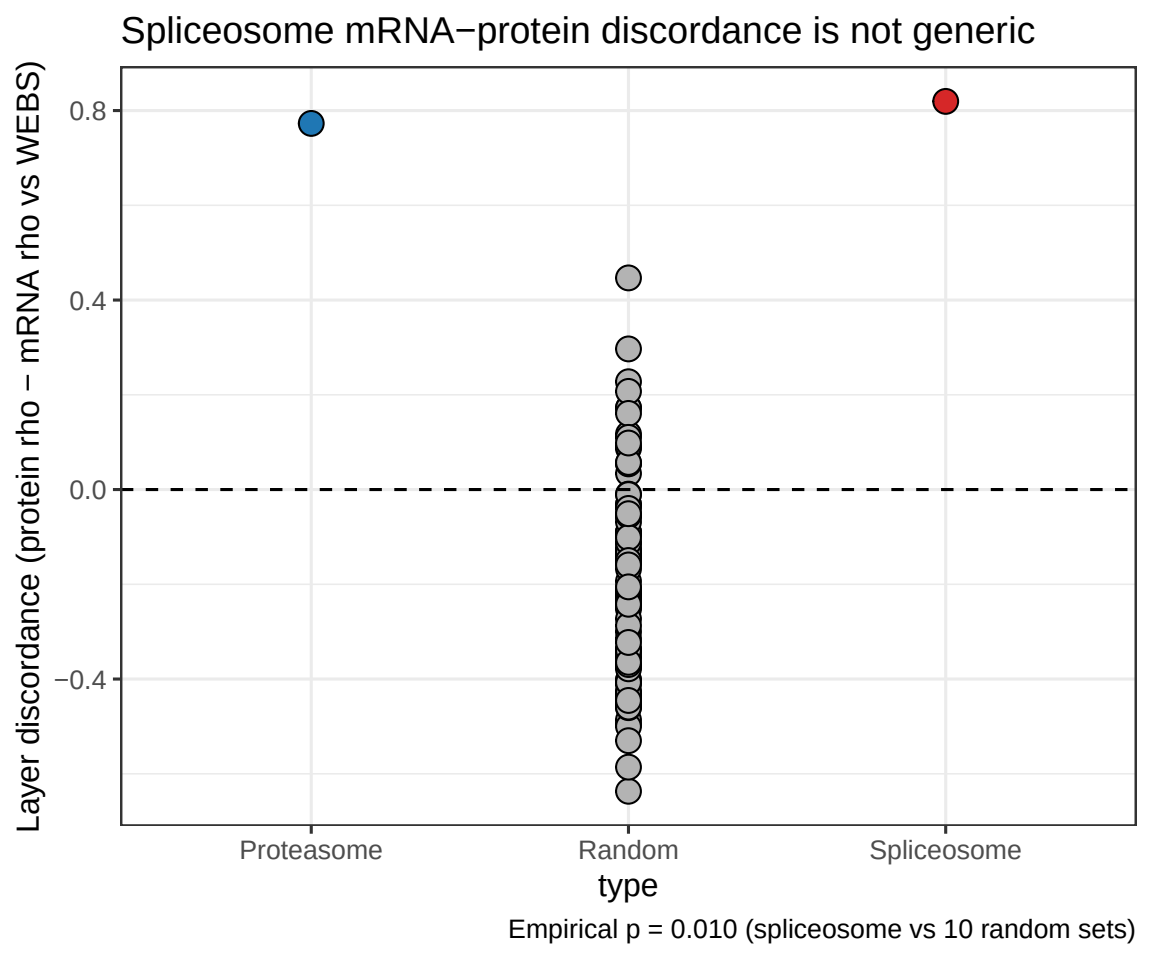

TCGA−BRCA
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.59
0
100
200
300
Time (months)


TCGA−SKCM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.8
0
100
200
300
400
Time (months)


Computational scalability of WEBS and alternatives
Runtime, seconds (log scale)
100.0
10.0
1.0
0.1
1000
3000
10000
30000
Number of cells (log scale)
Method
AUCell
MeanExpr
ssGSEA
WEBS


(a) WEBS sensitivity to maxRank (fixed at 1500 = 0.075)
Spearman rho vs default WEBS
1.0
0.9
0.8
0.7
0.6
0.05
0.10
0.15
0.20
0.25
0.30
0.50
0.75
0.90
UCell maxRank (fraction of genes)
(b) Leave−one/two−out robustness (n = 57)
Count
10
5
0
0.75
0.80
0.85
0.90
0.95
1.00
Spearman rho vs default WEBS
type
leave−one−out
leave−two−out
(c) Cross−platform (GSE131928):
patient−level WEBS rho = 0.67
(Smart−seq2 vs 10x, n = 10)


Feature importance by risk group (SHAP)
HDAC1
ITGA5
SIRT2
HIF1A
MMP2
SPP1
0.0
0.1
0.2
0.3
Mean |SHAP|
Risk group
High
Low
SHAP distribution per gene
SPP1
SIRT2
MMP2
ITGA5
HIF1A
HDAC1
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on risk score)
Risk group
High
Low


Spliceosome mRNA−protein discordance is not generic
Layer discordance (protein rho − mRNA rho vs WEBS)
0.8
0.4
0.0
−0.4
Proteasome
Random
Spliceosome
type
Empirical p = 0.010 (spliceosome vs 10 random sets)